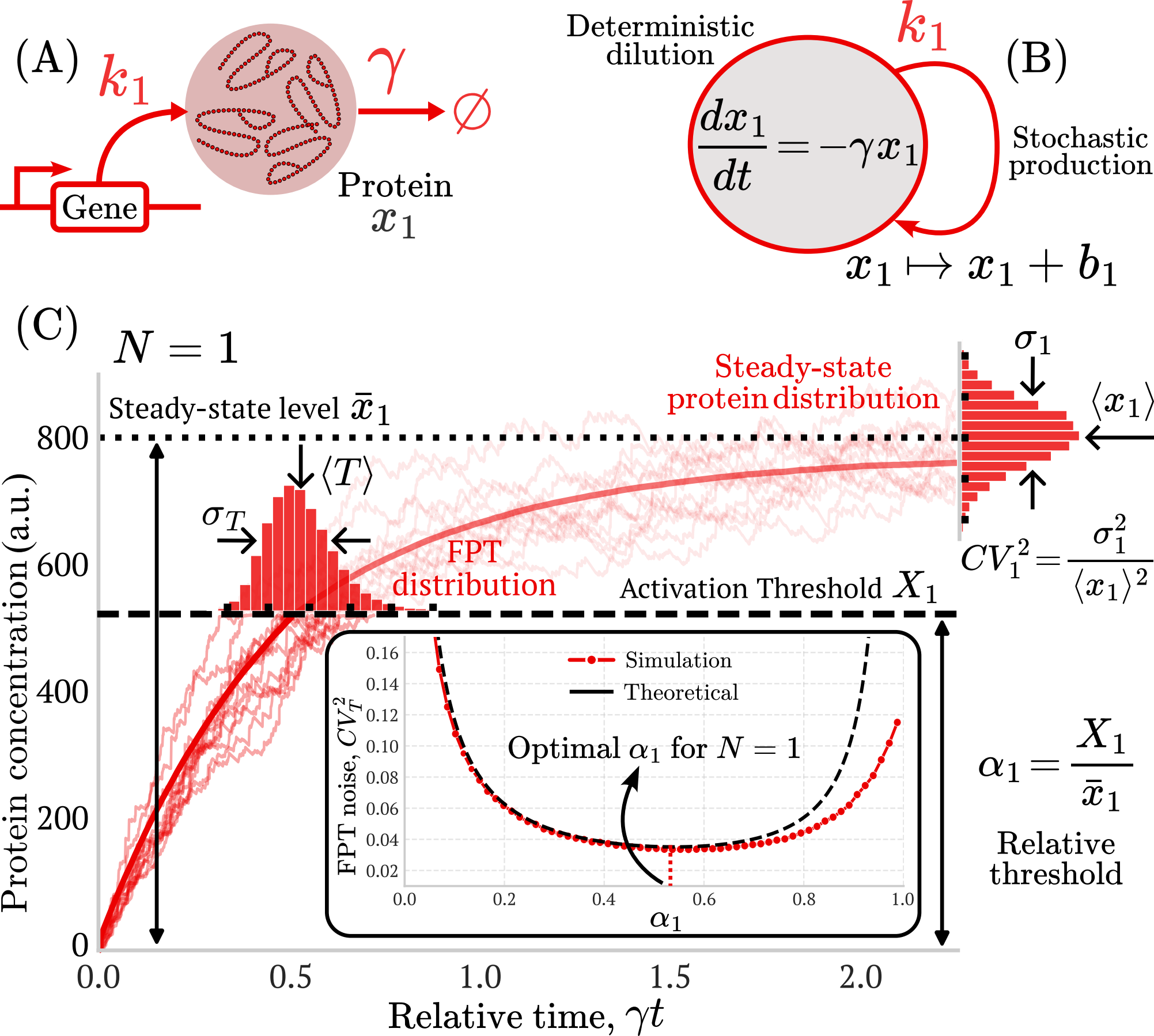
(A)
k_1
γ
$\emptyset$
Gene
Protein
x_1
Deterministic
dilution
k_1
(B)
$\frac{dx_1}{dt} = -\gamma x_1$
Stochastic
production
$x_1 \mapsto x_1 + b_1$
(C)
$N = 1$
Steady-state level $\bar{x}_1$
Steady-state
protein distribution
σ_1
$\langle x_1 \rangle$
$\langle T \rangle$
σ_T
FPT
distribution
Activation Threshold X_1
$CV_1^2 = \frac{\sigma_1^2}{\langle x_1 \rangle^2}$
Protein concentration (a.u.)
800
600
400
200
0
0.0
0.5
1.0
1.5
2.0
Relative time, γt
Simulation
Theoretical
FPT noise, CV_T^2
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.0
0.2
0.4
0.6
0.8
1.0
α_1
Optimal α_1 for $N = 1$
$\alpha_1 = \frac{X_1}{\bar{x}_1}$
Relative
threshold